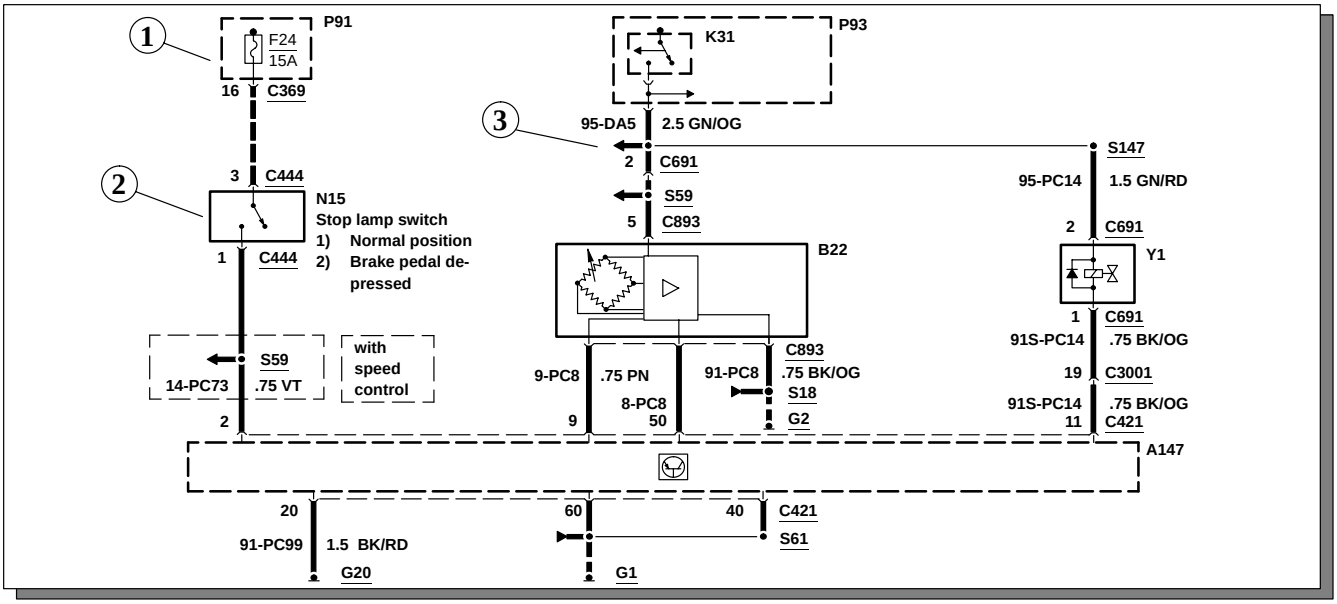
Note

All wiring connections between components are shown exactly as they exist in the vehicles. It is important to realize, however, that no attempt has been made on the schematic to represent components and wiring as they physically appear on the vehicle. For example, a 4-foot length of wire is treated no differently in a schematic from one which is only a few inches long. Furthermore, to aid in understanding electrical (electronic) operation, wiring inside complicated components has been simplified.

Complete Circuit Operation -1

Each circuit is shown completely and independently in one chapter or “cell”. Other components which are connected to the circuit may not be shown unless they influence the circuit operation.



Current Flow (1)

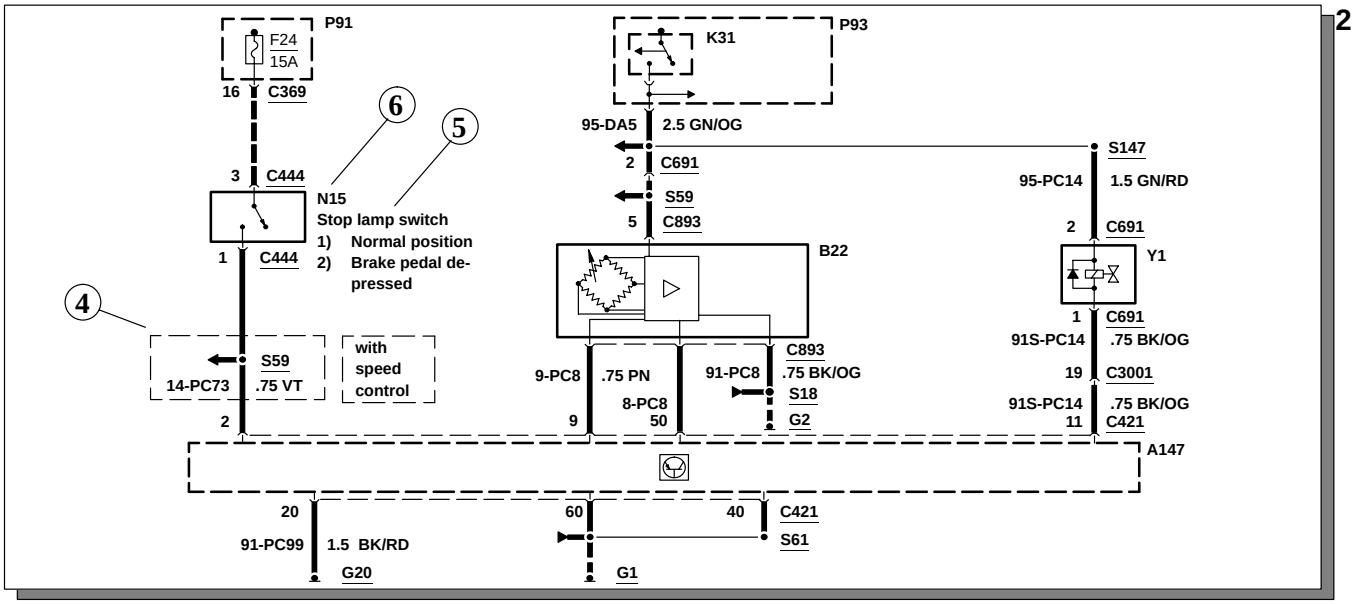
Each cell normally starts with the component that powers the circuit such as a fuse or the ignition switch. Current flow is shown from the power source at the top of the page to ground at the bottom of the page. In order to concentrate on the essential parts, power supply and ground connections are sometimes simplified by a dashed line in the schematics. A full representation of the power supply of a fuse or the power distribution from a fuse to various components is given in cells 700-02-00 “Power Distribution” and 700-03-00 “Fuse Details”, respectively. Full representation of the ground connections are shown in cell 700-04-00 “Ground Distribution”.

Switch Positions (2)

Within the schematic, all switches, sensors and relays are shown “at rest” (as if the Ignition Switch were OFF).

Splices (3)

An arrow indicates that the splice is not shown completely. The page where the splice appears in full is listed in the Index. If the splice is not shown completely in the schematic but is complete within the cell, the number of the page where the splice is completed is indicated next to the arrow.



Boxes (4)

A thin dashed box on a schematic indicates a part of the circuit which is only present for a particular vehicle model, country, or option. These qualifiers are shown next to the box on the schematic.

Component Names and Notes (5)

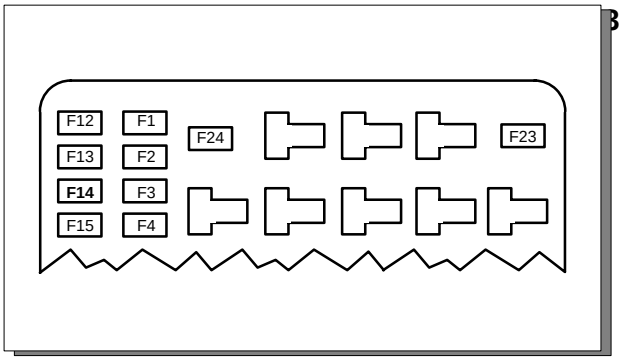
Component names are placed on the right hand side of each component. Any notes that describe switch positions or operating conditions follow the name. Descriptions of the internals of the component are also included here. The page where the component appears in full is listed in the Index.

Component Identification Numbers (6)

Each component on each schematic has a component identification number located to the upper right hand side of the component. By finding this number in the Component Location Chart, the Component Location View for that component can easily be found.

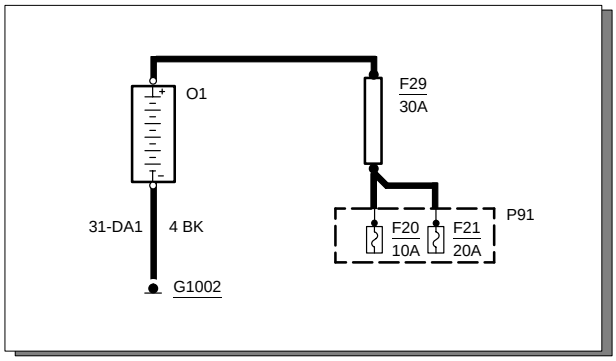
Fuse and Relay Information

Cell 700-01-00 “Fuse/Relay Information” contains a view of the fuse-/relay box in which all fuses and relays are identified. Cell 700-01-00 also specifies in table format which circuits/systems are protected by each fuse.



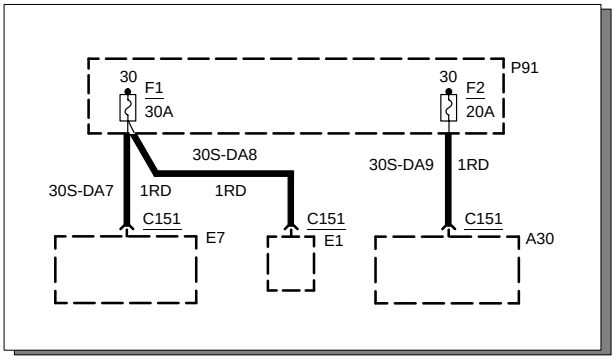
Power Distribution

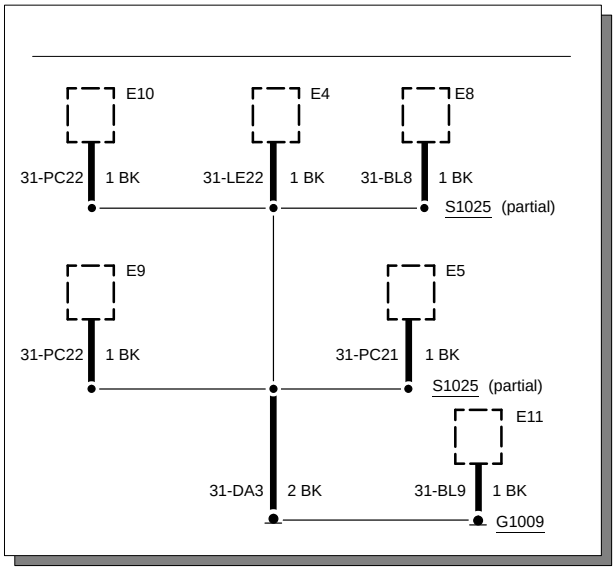
Cell 700-02-00 “Power Distribution” shows the current feed circuit. The current path is shown from the battery to the ignition switch and to all fuses.



Fuse Details

Cell 700-03-00 “Fuse Details” shows the circuits protected by each fuse. The circuit is traced from the fuse to the component. All details (wires, splices, connectors) between the fuse and the first component are shown.

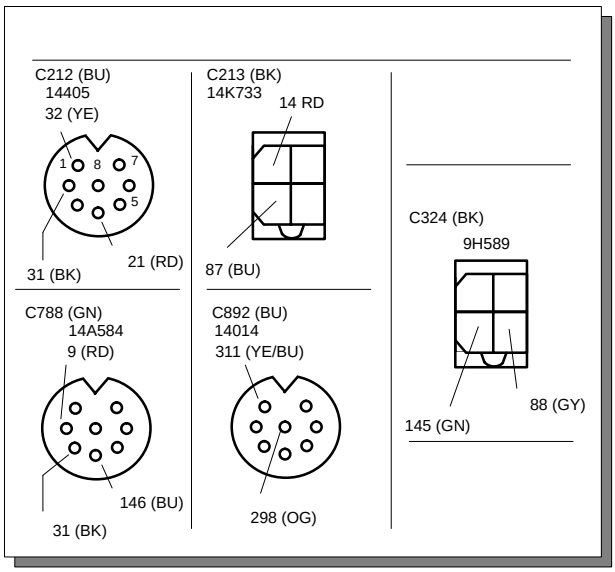




Ground Distribution -4

Cell 700-04-00 “Ground Disribution” contains the schematics that show the complete details for each ground connection or main ground splice. This is useful in diagnosing a problem affecting several components at once (poor ground connection or ground splice). All details (wires, splices, connectors) between the ground point and the components are shown. These ground connection details are shown here in order to keep the individual cell schematics as uncluttered as possible.

Splices with a large number of wires may be split in order to improve the graphic representation. A thin line is used to indicate the unity of the splice. The splice is then marked with the additional text “partial”.



Component Identification Information

Component Identification Information begins in cell 700-05-00 and consists of
(1) Component Location Chart (List),
(2) Component Location Views, and
(3) Connector Views (Faces).
The **Component Location Chart** helps the user find where the various items depicted on the schematic can physically be found on the vehicle. A brief written description of the location is given, along with a reference to the component location views.
The **Component Location Views** show the components and their connecting wires as they can be found on the vehicle.
The **Connector Views** show the views of the pins and/or cavities of all connectors. The pin and cavity sides are shown separately as if the connector were disconnected. The color of the connector housing is indicated next to the connector number. The harness causal number is located above the connector view and below the connector number. The wiring harness designations are listed in cell 700-05-00 “Component Location Charts”.

WARNINGS

- *Always wear safety glasses for eye protection.*
- *Use safety stands whenever a procedure requires being under a vehicle.*
- *Be sure that the* **Ignition Switch** *is always in the OFF position, unless otherwise required by the procedure.*
- *Set the parking brake when working on any vehicle. An automatic transmission should be in PARK. A manual transmission should be in NEUTRAL.*
- *Operate the engine only in a well-ventilated area to avoid danger of carbon monoxide.*
- *Keep away from moving parts, especially the fan and belts, when the engine is running.*
- *To prevent serious burns, avoid contact with hot metal parts such as the radiator, exhaust manifold, tail pipe, catalytic converter and muffler.*
- *Do not allow flame or sparks near the battery. Gases are always present in and around the battery cell. An explosion could occur.*
- *Do not smoke when working on a vehicle.*
- *To avoid injury, always remove rings, watches, loose hanging jewelry and avoid wearing loose clothing.*

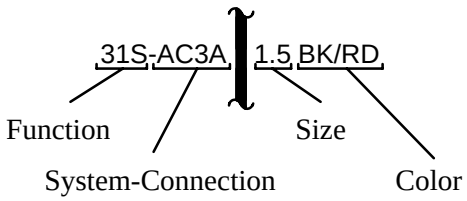
Circuit Numbering and Wire Identification

Circuit Numbering:

Ford has introduced a world-wide uniform system for circuit numbering and wire identification. The system is called Function, System-Connection, or FSC for short.

FSC was developed mainly to assist in vehicle development and production processes, but is also helpful in troubleshooting electrical circuits.

Example:



Function:

The first two digits identify the function of the wire. In this case, the function code includes the letter “S” to indicate it is an additionally switched function. The function part of the circuit number is the most helpful to the technician in troubleshooting a circuit. A complete list of function codes and their descriptions is included in cell 100-00-04 “Symbols”.

System-Connection (including branch):

Systems are related to subsets of the vehicle circuitry. A complete list of the system codes can be found in cell 100-00-04 “Symbols”.

Immediately following the system letter code is the connection number specific to that system. A branch identification is used to differentiate wires in one connection with the same function.

Wire Identification (wire colors): -6

The wire identification consists of a basic color and an identification color, and is determined directly from the wire’s circuit number. In the schematics the wire colors are indicated next to the wires.

The basic and identification colors are abbreviated using the international norm IEC 757. The abbreviations are listed in cell 100-00-04 “Symbols”.

Each function code has a specific basic color associated with it which can also be seen in the chart of function codes in cell 100-00-04 “Symbols”. The identification color is a colored stripe used to differentiate several wires with the same function within one component connector.

Using the circuit number and wire identification used in the above example we know the following information about the wire:

Function:	31 = Ground S = Additionally switched circuit
System:	AC = Headlamp leveling
Connection:	3 = Switch connection A = Branch
Size:	1.5 = 1.5 mm^2
Color:	BK = Basic color black (determined by function 31) RD = Identification color red (red stripe)